Tag name is DWELL *using Data Type* **FBD_TIMER**.

Format when dragged in:
```
TONR( )
```

Timer structure
```
DWELL.PRE := 5000;
DWELL.TimerEnable := SWITCH;
TONR(DWELL);
```

Timer Values	Timer status bit formats
DWELL.PRE	DWELL.EN
DWELL.ACC	DWELL.TT
	DWELL.DN

Examples:

Energize Timer Delay ON
```
DWELL.PRE := 5000;
DWELL.TimerEnable:= SWITCH;
TONR(DWELL);
OUTPUT := DWELL.DN;
```

Energize Timer Delay OFF
```
DWELL.PRE := 5000;
DWELL.TimerEnable:= SWITCH;
TONR(DWELL);
OUTPUT := NOT DWELL.DN AND DWELL.EN;
```

Setting Preset
```
DWELL.PRE  := TIME_VALUE;
```

Energize Timer Delay ON
```
DWELL.PRE := 5000;
DWELL.TimerEnable := SWITCH;
TONR(DWELL);
IF DWELL.DN THEN
     OUTPUT := 1;
     additional actions
ELSE;
     OUTPUT := 0;
     additional actions
END_IF;
```

Energize Timer Delay OFF
```
DWELL.PRE:= 5000;
DWELL.TimerEnable := SWITCH;
TONR(DWELL);
IF NOT DWELL.DN AND DWELL.EN THEN
     OUTPUT := 1;
ELSE;
     OUTPUT := 0;
END_IF;
```

Tag name ONE_SHOT *using Data Type* **FBD_ONESHOT**.

One Shot Rising structure
```
ONE_SHOT.InputBit := PUSHBUTTON;
OSRI(ONE_SHOT);
```

Format when dragged in:
```
OSRI( )
```

Examples:

Format for assigning Tag to Output Bit:
```
ONE_SHOT.InputBit := PUSHBUTTON;
OSRI(ONE_SHOT);
MY_TAG := ONE_SHOT.OutputBit
```

One shot bit used with additional conditions
```
ONE_SHOT.InputBit := PUSHBUTTON;
OSRI(ONE_SHOT);
IF ONE_SHOT.OutputBit AND OK_TO_RUN THEN
MY_TAG := 1;
ELSE;
MY_TAG := 0;
```

A dialog opens when the TON *instruction is dragged into the program, the tag name in this example is* DWELL. *The* TON *is a System block and Program resource.*

Timer structure

```
"DWELL".TON(IN:= "SWITCH",
            PT:=T#5000ms);
```

Timer values	Timer status bit formats
`"DWELL".PT`	`"DWELL".Q`
`"DWELL".ET`	`"DWELL".IN`

Format when dragged in:

```
"DWELL".TON(IN:=_bool_in_,
            PT:=_time_in_,
            Q=>_bool_out_,
            ET=>_time_out_);
```

The IN and PT are required.

If the Q and ET are unassigned, they will be deleted when pressing ENTER to complete the instruction.

Examples:

Energize Timer Delay ON

```
"DWELL".TON(IN:= "SWITCH",
            PT:=T#5000ms);
"OUTPUT" := NOT "DWELL".Q;
```

Energize Timer Delay OFF

```
"DWELL".TON(IN:= "SWITCH",
            PT:=T#5000ms);
"OUTPUT" := NOT "DWELL".Q AND "DWELL".IN;
```

Setting Prest Time

```
"DWELL".PT := T#5000ms;
```

Energize Timer Delay ON

```
"DWELL".TON(IN:= "SWITCH",
            PT:=T#5000ms);
IF "DWELL".Q THEN
   "OUTPUT" := 1;
    additional actions
ELSE;
   "OUTPUT" := 0;
    additional actions
END_IF;
```

Energize Timer Delay OFF

```
"DWELL".TON(IN:= "SWITCH",
           PT:=T#5000ms);
IF NOT "DWELL".Q AND "DWELL".IN THEN
   "OUTPUT" := 1;
ELSE;
   "OUTPUT" := 0;
END_IF;
```

A dialog opens when the R_TRIG *instruction is dragged into the program, the tag name in this example is* ONE_SHOT. *The* R_TRIG *is a System block and Program resource.*

```
//#PULSE is a Local Instance.
```

One Shot Rising structure

```
"ONE_SHOT"(CLK:="PUSHBUTTON",
                  Q=>#PULSE);
 MY_TAG := #PULSE
```

Format when dragged in:

```
"ONE_SHOT"(CLK:=_bool_in,
            Q=>_bool_out_);
```

Notice the instruction name is not shown.

One shot bit used with additional conditions

```
"ONE_SHOT"(CLK:="PUSHBUTTON",
                  Q=>#PULSE);
IF #PULSE AND OK_TO_RUN THEN
    MY_TAG := 1;
ELSE
    MY_TAG := 0;
END_IF;
```